A Fine-Tuned Small Language Model for Interpreting Fluid–Structure Interaction Experiments

PIV experiment · U-Net flow surrogate · fine-tuned small language model · retrieval grounding · REST and MCP

***Kumar, Angad**
angadkumar16ak@gmail.com
Obra Tompkins High School, Texas, USA

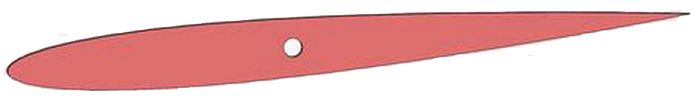
Shukla, Khemraj
khemraj_shukla@brown.edu
Brown University, Rhode Island, USA

**presenting author*

CODE & DATA
github.com/AngadKumar16/FSI-SLM
REST API · MCP server · weights

One laptop predicts the flow around a wing and explains what the measurement means — work that used to need a cluster and a specialist.

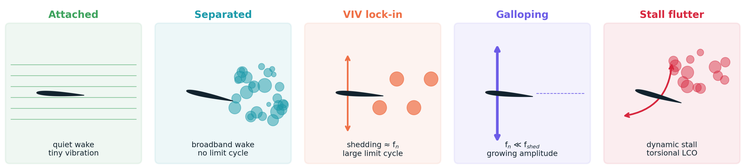
MOTIVATION



The specimen. A NACA 0009 section, 9% thick, pinned through the hole. Every measured number on this poster comes from this part.

- Moving fluid shakes structures. Enough shaking breaks them.
- A **PIV** camera answers this with thousands of velocity vectors per frame.
- Turning those vectors into a verdict takes an expert. **Experts are the bottleneck.**
- A frontier chatbot talks about the physics, invents the rest, and needs a data centre.

Five fluid-structure interaction regimes the model distinguishes



F1. The call the expert makes. Only the last three are dangerous. This is the judgement being automated.

WHAT WE BUILT

1. One **JSON record** for any run — measured or simulated.
2. A **small** language model, fine-tuned to read that record.
3. **Retrieval** from published papers, so it cites instead of inventing.
4. A network that predicts flow **and** structure from shape alone, replacing the solver.
5. All of it on **one laptop**, reachable over REST and MCP.

DATA

- 107 airfoil sections × 18 angles at Re = 5×10^5. **1,928 cases solved.**
- Three solvers, one output format: panel 0.13 s, RANS 4 min, URANS hours.
- A beam model turns surface pressure into deflection, so every case carries structure.
- **NACA 0009 never appears in training** — every number quoted for it is generalisation.

REFERENCES

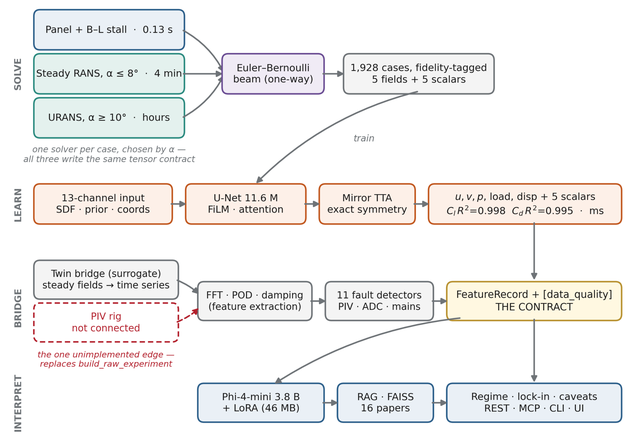
Ronneberger, O. et al. (2015) U-Net. MICCAI. · Thuerey, N. et al. (2020) Deep learning methods for RANS simulations of airfoil flows. AIAA J. 58(1).
Lewis, P. et al. (2020) Retrieval-augmented generation for knowledge-intensive NLP tasks. NeurIPS. · Leishman & Beddoes (1989) A semi-empirical model for dynamic stall. JAHS 34(3).
Hess & Smith (1967) Calculation of potential flow about arbitrary bodies. Prog. Aerospace Sci. 8. · Abbott & von Doenhoff (1959) Theory of Wing Sections. Dover. · Anthropic (2024) Model Context Protocol specification.

ACKNOWLEDGEMENT

With thanks to the AGU Bright STaRS program. Built with Python, PyTorch, OpenFOAM, MLX and FAISS. Code, data and weights: **github.com/AngadKumar16/FSI-SLM**

METHODS

- The network reads **13 channels**: the shape, the flow conditions, and the ideal-flow solution on the same grid.
- So it learns the **correction** to ideal flow, not the flow itself. 11.59 M parameters.
- The reader is **Phi-4-mini** (3.8 B) with a 46 MB LoRA adapter.
- Answers are grounded by retrieval: 16 papers, 1,840 passages, top three above a 0.30 score enter the prompt.
- **Eleven fault detectors** label dropout, clipping and mains hum before the model sees anything.
- Served three ways from one laptop — CLI, **FastAPI REST**, and an **MCP** server of twelve tools an agent can drive.

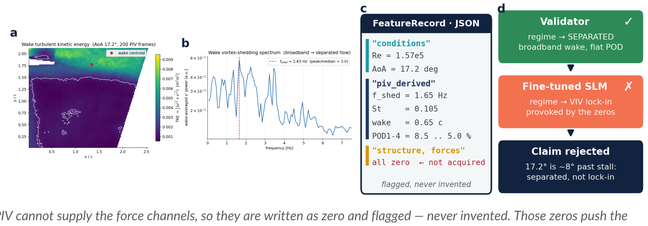


*F2. The twin, end to end. **The dashed red edge is the one link not yet built** — F3 is the measured path running beside it.*

A REAL MEASUREMENT

200 PIV frames of the specimen at **α = 17.2°**, pushed through the **same unchanged code** as the simulations.

Real PIV, end to end: 200 measured frames, one record, one verdict



*F3. PIV cannot supply the force channels, so they are written as zero and flagged — never invented. Those zeros push the language model into a wrong lock-in call, and the rule-based validator **overrules it**.*

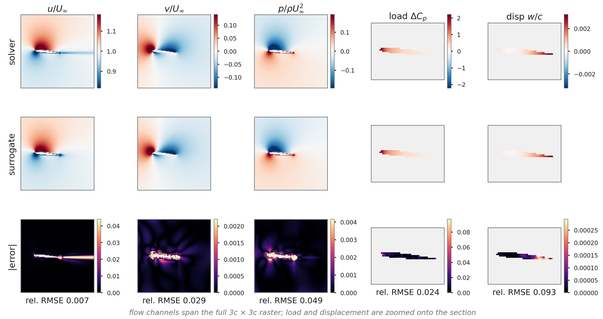
WHY TWO KINDS OF SOLVER

- Cheap solves are fast and cover every angle. Expensive solves are right and rare.
- Train on cheap alone: the **flow** comes out right, the **structure** does not.
- The beam amplifies the leading-edge pressure difference — exactly where the two solvers disagree.
- **47 expensive cases close most of that gap.** Cheap buys coverage; expensive buys truth.

RESULTS

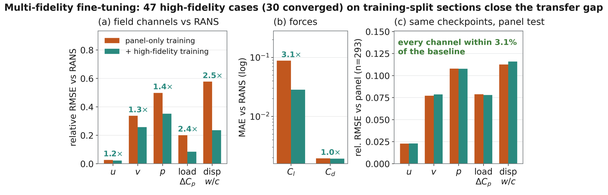
R^2 = 0.998 lift, on shapes it has never seen	**2.3%** velocity-field error	**293** held-out cases, 17 unseen sections
0.86 lock-in F1 (recall 1.000)	**47** expensive cases close the gap	**1** laptop, no GPU anywhere

NACA 0009 (held out) at α = 5°, Re = 5×10^5

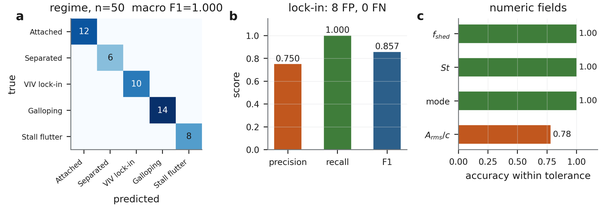


*F4. Solver, network and error on NACA 0009 at α = 5°, a section held out entirely. **Error concentrates in the near-wall layer and the wake**, exactly where it should. Relative RMSE 0.007 (u), 0.029 (v), 0.049 (p), 0.024 (load), 0.093 (deflection), against the panel targets the network trained on.*

Multi-fidelity fine-tuning: 47 high-fidelity cases (30 converged) on training-split sections close the transfer gap



*F5. The 47 expensive cases close the structural gap while every **field** channel stays within 3.1% of the cheap-only baseline. Honest cost: drag error regresses 0.00230 → 0.00314; lift error improves 11%.*



*F6. The reader on 50 held-out records. Regime classification is **saturated at this sample size**, so the metric that discriminates is lock-in detection: it never misses one and over-calls eight — a bias inherited from the rule-based labeller.*

SUMMARY

- Predicts velocity, pressure, load and deflection on shapes it has never seen.
- Reads the record, names the regime, flags faults, cites its sources.
- On the real run the validator **caught the model over-reaching**.
- Both halves run on one laptop, over REST and MCP.

FUTURE WORK

- Measure deflection alongside PIV, so real records stop carrying zeros.
- Finish the URANS sweep to make the deep-stall branch trainable.
- Close the loop: let deflection feed back into the flow.
- Calibrated uncertainty, and a harder interpreter test set.